Sec: SR.IIT_*COSC(MODEL-A) | GTM-3 | Date: 23-11-24
Time: 3 Hrs | JEE-MAIN | Max. Marks: 300

KEY SHEET

PHYSICS

1	**B**	2	**C**	3	**A**	4	**B**	5	**C**
6	**B**	7	**D**	8	**B**	9	**C**	10	**B**
11	**D**	12	**A**	13	**B**	14	**D**	15	**B**
16	**A**	17	**B**	18	**D**	19	**A**	20	**B**
21	**6**	22	**101**	23	**4**	24	**5**	25	**4000**

CHEMISTRY

26	**D**	27	**C**	28	**A**	29	**C**	30	**C**
31	**B**	32	**C**	33	**C**	34	**B**	35	**C**
36	**B**	37	**C**	38	**D**	39	**B**	40	**B**
41	**A**	42	**A**	43	**D**	44	**B**	45	**D**
46	**280**	47	**24**	48	**4**	49	**5**	50	**3**

MATHEMATICS

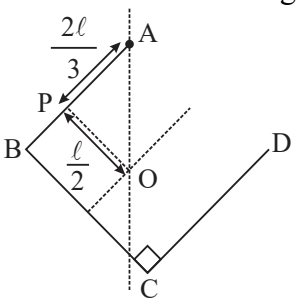
51	**D**	52	**C**	53	**D**	54	**A**	55	**B**
56	**D**	57	**A**	58	**C**	59	**D**	60	**B**
61	**A**	62	**D**	63	**D**	64	**C**	65	**A**
66	**C**	67	**A**	68	**B**	69	**B**	70	**C**
71	**2**	72	**11**	73	**0**	74	**8**	75	**5**

For More Join: @IITJEE_Advanced

SOLUTIONS

PHYSICS

1. Vertical line from hinge A must pass through C.M. of rod system.

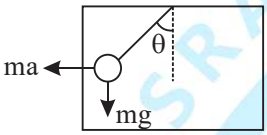


$$\tan\theta = \frac{OP}{AP} = \frac{\ell/2}{2\ell/3}$$

$$\tan\theta = \frac{3}{4} \Rightarrow \theta = \tan^{-1}\left(\frac{3}{4}\right)$$

2. By work energy theorem from trolley

$maL\sin\theta = mgL(1-\cos\theta)$



$$\Rightarrow \quad \tan\frac{\theta}{2} = \frac{a}{g}$$

$$\Rightarrow \quad \theta = 2\tan^{-1}\left(\frac{a}{g}\right)$$

3. $x = 2t \Rightarrow v_x = \frac{dx}{dt} = 2$

$$y = 2t^2 \Rightarrow v_y = \frac{dy}{dt} = 4t$$

$$\therefore \quad \tan\theta = \frac{v_y}{v_x} = \frac{4t}{2} = 2t$$

Differentiating with respect to time we get,

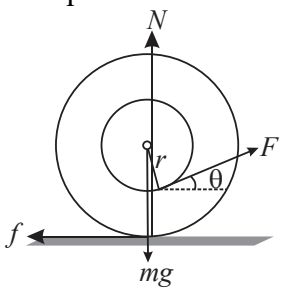
$$\left(\sec^2\theta\right)\frac{d\theta}{dt} = 2$$

$$\Rightarrow \quad \left(1+\tan^2\theta\right)\frac{d\theta}{dt} = 2 \Rightarrow \left(1+4t^2\right)\frac{d\theta}{dt} = 2$$

$$\Rightarrow \quad \frac{d\theta}{dt} = \frac{2}{1+4t^2}$$

$$\Rightarrow \quad \left[\frac{d\theta}{dt}\right]_{t=2} = \frac{2}{1+4(2)^2} = \frac{2}{17}\text{ rad/s}$$

4. For spool to be stationary, $\Sigma F = 0$



or $F\cos\theta - f = 0$ …(i)

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Also, $\Sigma\tau = 0$,

or $Fr - fR = 0$ …(ii)

From above equations,

$\cos\theta = \frac{r}{R}$.

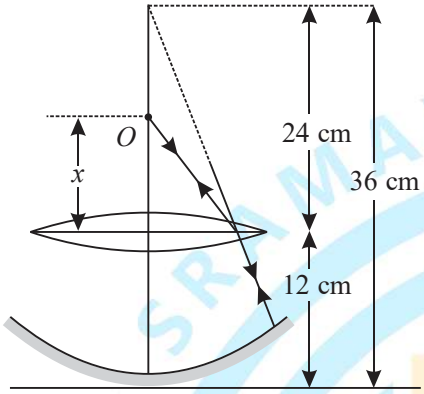
5. Covered area AB on earth $= 2\pi R^2(1-\sin\theta)$

[Solid angle $= 2\pi(1-\sin\theta)$]

Area on earth escaped from satellite

$= 4\pi R^2 - 2\pi R^2(1-\sin\theta) = 2\pi R^2(1+\sin\theta)$

6. $v = 24\,cm, u = x$



Using

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

or

$\frac{1}{-24} - \frac{1}{(-x)} = \frac{1}{40}$

$\therefore \quad x = 15\,\text{cm}$

7. The buoyant force,

$F_b = P_{\text{bottom}}\,A - P_{\text{top}}\,A$

$\Rightarrow \quad P_{\text{bottom}}\,A = F_b + P_{\text{top}}\,A$

$= V\rho g + \rho g h \times \pi R^2$.

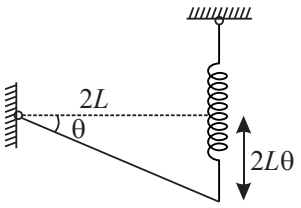
8. $0 = m_1 v_1 - m_2 v_2$

and $\frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2 = E$

After simplifying above equations, we get

$v_1 + v_2 = \sqrt{\frac{2E(m_1+m_2)}{m_1 m_2}}$

9.



The restoring torque for small θ

$\tau_{\text{nest}} = -[k(2L\theta)\times 2L]$

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and $\alpha = \frac{\tau_{rest}}{I}$

$= \frac{4kL^2}{\frac{M(2L)^2}{3}}(-\theta) = \frac{3k}{M}(-\theta)$

On comparing with $\alpha = -\omega^2\theta$, we get

$\omega = \sqrt{\frac{3k}{M}}$.

10. Charge of the whole sphere, $Q = \left(\frac{4}{3}\pi R^3\right)\rho$

Charge of the cavity, $q = \frac{4}{3}\pi\left(\frac{R}{2}\right)^2\rho = \frac{Q}{8}$

Now,

$V_o = \left[\left(V_{\text{whole sphere}}\right)_o - \left(V_{\text{cavity}}\right)_o\right]$

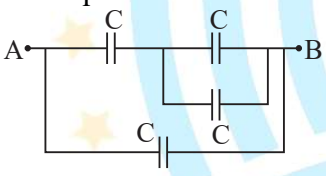
$= \frac{3}{8\pi\varepsilon_0}\frac{Q}{R} - \frac{1}{4\pi\varepsilon_0}\frac{\left(\frac{Q}{8}\right)}{\frac{R}{2}}$

$= \frac{5}{16\pi\varepsilon_0}\frac{Q}{R}$

$= \frac{5}{16\pi\varepsilon_0 R}\times\left(\frac{4}{3}\pi R^3\rho\right)$

$= \frac{5}{12}\frac{R^2\rho}{\varepsilon_0}$.

11. The equivalent circuit is shown in figure



Thus $C_{eq} = \frac{5C}{3} = \frac{5\varepsilon_0 A}{3d}$.

12. 25% total energy $= 150\times10^{15}\times\frac{1}{4}MeV = 6kW$

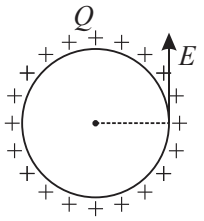
13. Induced electric field

$E = \frac{r}{2}\left(\frac{dB}{dt}\right)$

$= \frac{rR}{2}$.

This electric field exerts tangentive force on each part of the ring. So net force,

$F = EQ = \frac{rRQ}{2}$.



For ring to be in equilibrium,

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$F \le F_{\text{friction}}$

or $r\dfrac{RQ}{2} \le \mu mg$

$\therefore \quad \mu \ge \dfrac{rRQ}{2mg}$

14. $n_1 + n_2 = n_{\text{mix}}$

$\dfrac{P_1V}{T_1} + \dfrac{P_2V}{T_2} = \dfrac{P(2V)}{T}$

$\Rightarrow \dfrac{P}{T} = \dfrac{1}{2}\left(\dfrac{P_1}{T_1} + \dfrac{P_2}{T_2}\right)$

Also $U_1 + U_2 = U_{\text{mix}}$

$\Rightarrow \dfrac{5}{2}P_1V + \dfrac{5}{2}P_2V = \dfrac{5}{2}P(2V)$

$\Rightarrow P = \dfrac{P_1 + P_2}{2}$

15. $\Delta x_1 = (\mu_1 - 1)t = (1.5-1)t = 0.5t$ (upward)

and $\Delta x_2 = (\mu_2 - 1)\times 2t = \left(\dfrac{4}{3} - 1\right)\times 2t = \dfrac{2}{3}t$ (downward).

As $\Delta x_2 > \Delta x_1$, so shift will be along -ve y-axis.

16. $\dfrac{dk}{dx} = \dfrac{dU}{dx}$ (at all points)

$V_p = -(V)\left(\dfrac{dy}{dx}\right)$

17. D_2 is forward Biased and D_1 is reverse Biased.

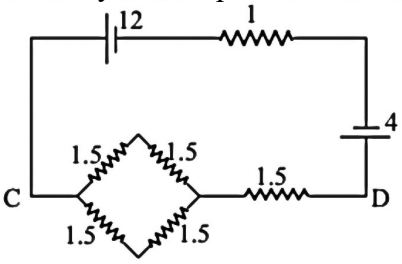
18. Rate of cooling $\dfrac{\Delta\theta}{t} = \dfrac{A\varepsilon\sigma\left(T^4 - T_0^4\right)}{mc}$

$\Rightarrow t \propto \dfrac{m}{A}\left[\because \Delta\theta, t, \sigma, \left(T^4 - T_0^4\right) \text{ are constant }\right]$

$\Rightarrow t \propto \dfrac{m}{A} \propto \dfrac{\text{Volume}}{\text{Area}} \propto \dfrac{a^3}{a^2} \Rightarrow t \propto a \Rightarrow \dfrac{t_1}{t_2} = \dfrac{a_1}{a_2}$

$\Rightarrow \dfrac{100}{t_2} = \dfrac{1}{2} \Rightarrow t_2 = 200\text{sec.}$

19. In steady state capacitor acts as open circuit,



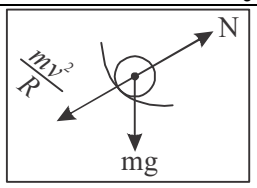
$R_{eq} = R_{eq}\text{of} = 4$

$V_{CD} = \left[\dfrac{(12-4)}{4}\right]3 = 6$

$\therefore V_{6\mu F} = \dfrac{3\times 6}{3+6} = 2$

20. By energy conservation, we have

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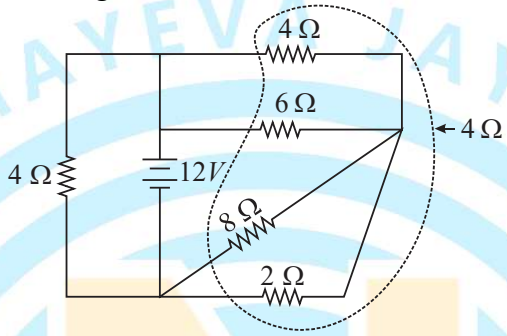
$$\frac{1}{2}mv^2 = mgR\sin\theta$$

$$\Rightarrow \quad \frac{mv^2}{R} = 2mg\sin\theta$$

$$N = \frac{mv^2}{R} + mg\sin\theta$$

$$N = 3mg\sin\theta$$

21. The effective circuit is shown in figure.



The equivalent resistance, $R = 2\Omega$

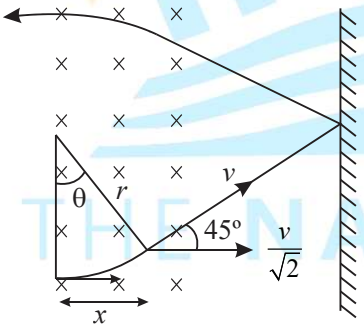
$\therefore$ Current $\quad i = \frac{12}{2} = 6\,\text{A}$

22. $\frac{2T}{r} = \frac{2\times 0.07}{0.14\times 10^{-3}} = 10^3\ \text{N/m}^2$

Pressure applied, $= P_a + \frac{2T}{r}$

$= 10^5 + 10^3 = 101\times 10^3\ N/m^2$

23. $r = \frac{mv}{qB}$



$$\sin\theta = \frac{x}{r}$$

$$= \frac{\frac{mv}{\sqrt{2}qB}}{\frac{mv}{qB}} = \frac{1}{\sqrt{2}}$$

or $\quad \theta = \frac{\pi}{4}$

Time to complete the circle $(2\pi), T = \frac{2\pi m}{qB}$

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$\therefore$ time taken to traverses $\frac{\pi}{4}, t = \frac{\pi m}{4qB}$

Time taken to travel horizontal distance

$$t_1 = \frac{\frac{mv}{\sqrt{2}qB}}{\frac{v}{\sqrt{2}}} = \frac{m}{qB}$$

Total time taken $= 2t + 2t_1$

$$= \frac{m}{2qB}(\pi + 4)$$

24. Horizontal acceleration of the system is

$$a = \frac{F}{2m + m + 2m} = \frac{F}{5m}$$

Let N be the normal reaction of the system is

$$N = 2ma = \frac{2F}{5}$$

Now B will slide downwards of

$$\mu N \geq m_B g$$

$$\mu\left(\frac{2F}{5}\right) \geq mg$$

$$\Rightarrow \quad F \geq \frac{5mg}{2\mu}$$

25. As heat produced in an $L-C-R$ circuit, i.e. Heat $= (i_{\text{rms}})^2 Rt$

The rms value of current $i_{rms} = \frac{i_0}{\sqrt{2}} = \frac{25}{Z\sqrt{2}}$

where, $Z = \sqrt{R^2 + (X_L - X_C)^2} = 5\Omega$

Heat $= \left(\frac{25}{5\sqrt{2}}\right)^2 4 \times 80 = 4000\,\text{J}$

CHEMISTRY

26. $K_P = K_C(RT)^{\Delta_{ng}}$

$\frac{K_P}{K_C} = 2 \Rightarrow (RT)^{\Delta_{ng}} = 2$

$(0.0821 \times \text{T})^{-2} = 2$

$\frac{1}{2} = (0.0821 \times \text{T})^2$

$T \simeq 8.61K$

27. Rate $= K[A][B]$

$2 \times 10^{-4} = K \times 1 \times 10^{-2} \times 2 \times 10^{-2}$

$\Rightarrow \quad K = 1$

Rate$\uparrow$ 4 time on doubling conc. of Both reactants

28. $3A + 2B^{+3} \rightarrow 2B + 3A^{+2}$

$$E_{cell} = E^o_{cell} - \frac{0.06}{6}\log\frac{[A^{+2}]^3}{[B^{+3}]^2}$$

$$\Rightarrow \quad E^o_{cell} - 0.01\log\frac{(10^{-2})^3}{(10^{-1})^2}$$

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$\Rightarrow \quad E^o_{cell} + 0.01 \times 4$

29. $\text{pH} = 6 = \text{pK}_a + \log\dfrac{0.1+x}{0.1-x}$

$\dfrac{0.1+x}{0.1-x} = 10 \Rightarrow x = 0.082$

$\text{Ca(OH)}_2 \rightarrow 0.041\text{M}$ Required

M moles = $0.041 \times 100 = 4.1$

$\text{Wt.} = \dfrac{4.1 \times 74}{1000} = 0.3g$

30. For $C_6H_{12}O_6$ %C = 40%

%H = 6.66%

%O = 53.33%

31. Eq. of mixture = $1 + 3 \times 3 + 4 \times 6 = 34$

Eq. of mixture = Eq. of $K_2Cr_2O_7$

$n = \dfrac{34}{6} = \dfrac{17}{3}$

32. At eq. $\Delta G = 0$

$\dfrac{\Delta H}{\Delta S} = T \Rightarrow \dfrac{200 \times 1000}{2000} = 100$

33. $[\text{Co(en)}_2(\text{NH}_3)_2]^{+3}$ is optically active

34. Max. atoms = 6

35. Mn^{3+} and Co^{3+} readily gain an electron to reach a more stable +2 oxidation stable.

36. $x = a_0 \times \dfrac{25}{3}$

$y = a_0 \times \dfrac{9}{4}$

$y = \dfrac{3x}{25} \times \dfrac{9}{4} = \dfrac{27x}{100}$

37. $4d \simeq 5d$ Radius

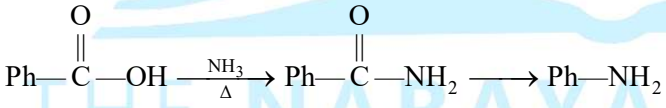
38. Highest complex forming tendency is of Lu^{+3}

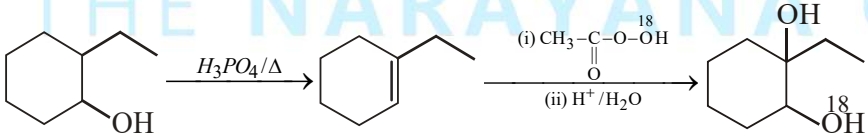
39. D.M. order : $H_2O > HF > NH_3 > H_2S$

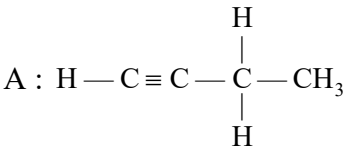
40. Wolff-Kishner reduction

41. Cellulose has β-D Glucose

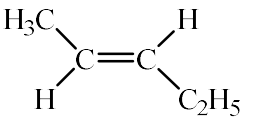
Unit and β-1, 4-glycosidic linkage.

42. 

43. 

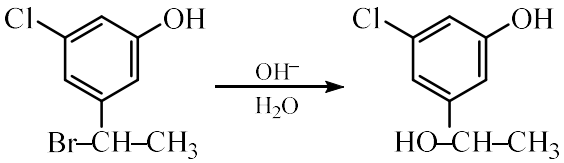
44. A : 

B : $CH_3 - C \equiv C - CH_2 - CH_3$

C : 

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45.



46. $200 = P_A^o \times \frac{1}{4} + P_B^o \times \frac{3}{4}$

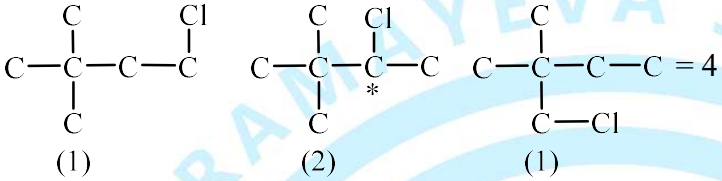
$300 = P_A^o \times \frac{3}{4} + P_B^o \times \frac{1}{4}$

$P_A^o = 350$

$P_s = 350 \times \frac{4}{5}$

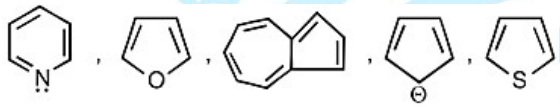
47. $\Delta H = 3\times110 + 3\times150 + 36 - 6\times100 - 3\times80 = -24$

48.



(1) (2) (1)

49.



50. A is $CH_2=\overset{H}{\overset{|}{C}}-\underset{H}{\overset{|}{C}}=\overset{H}{\overset{|}{C}}-\underset{H}{\overset{|}{C}}=CH_2$

MATHEMATICS

51. $\left(\frac{1}{a_1}-\frac{1}{a_2}\right)+\left(\frac{1}{a_2}-\frac{1}{a_3}\right)+\ldots\left(\frac{1}{a_9}-\frac{1}{a_{10}}\right)=\frac{9d}{64}$

$\Rightarrow \frac{a_{10}-a_1}{a_1a_{10}}=\frac{9d}{64}$

$\Rightarrow a_1a_{10}=64$

Also $(a_1+a_{10})\left(\frac{1}{a_1a_{10}}+\ldots\frac{1}{a_{10}a_1}\right)=\frac{a_1+a_{10}}{10}\left(\frac{1}{a_1}+\ldots\frac{1}{a_{10}}\right)$

$\Rightarrow 2\left(\frac{1}{a_1}+\ldots\frac{1}{a_{10}}\right)=\frac{a_1+a_{10}}{10}\left(\frac{1}{a_1}+\ldots\frac{1}{a_{10}}\right)$

$\Rightarrow a_1+a_{10}=20$

52. $\because$ α,β, are the roots of $x^2+px+q=0$

Then $\alpha+\beta=-p, \alpha\beta=q$

Also α,β are roots of $x^{3900}+p^{1950}x^{1950}+q^{1950}=0$

$\because$ $\alpha^{1950}+\beta^{1950}=-p^{1950}$ and $\alpha^{1950}\beta^{1950}=q^{1950}=0$

Now, α/β is a root of $x^n+1+(x+1)^n=0$

Then $\left(\frac{\alpha}{\beta}\right)^n+1+\left(\frac{\alpha}{\beta}+1\right)^n=0$

$\Rightarrow \alpha^n+\beta^n+(\alpha+\beta)^n=0$

$\Rightarrow \alpha^n+\beta^n+(-p)^n=0$

or $\alpha^n+\beta^n=-(-p)^n$

$(\because \alpha+\beta=-p)$ ……..(iii)

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From equation (ii) and (iii), we get $n = 1950$

53. $\dfrac{\sin 70^\circ - \sin 10^\circ}{\sin 10^\circ . \sin 70^\circ} + \dfrac{1}{\cos 40^\circ}$

$\dfrac{\cos^2 40^\circ + \sin 10^\circ \sin 70^\circ}{\sin 10^\circ \sin 50^\circ \sin 70^\circ}$

$8\left[\cos^2 40^\circ + \sin 10^\circ . \sin 70^\circ\right]$

$\Rightarrow 4\left[1 + \cos 80^\circ + \cos 60^\circ - \cos 80^\circ\right]$

$a = 6$

$b = \cos 18^0 - \dfrac{3}{\cos 18^0} - 2\tan 18^0$

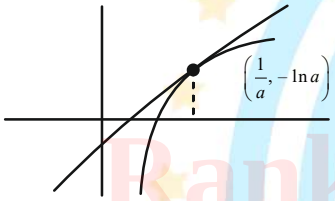
$= \dfrac{4\cos^2 18^0 - 3 - 2\sin 18^0}{\cos 18^0}$

$= \dfrac{2\left(1 + \cos 36^0\right) - 2\sin 18^0 - 3}{\cos 18^0}$

$= \dfrac{2\left(1 + \cos 36^0 - \sin 18^0\right) - 3}{\cos 18^0}$

$= \dfrac{2\left(1 + \dfrac{1}{2}\right) - 3}{\cos 18^0} = 0.$

54.



$f'(x) = \dfrac{1}{x_1}$, $f'(x) = a$, $a = \dfrac{1}{x_1}$

$f(x) = -1 + 1 + a = a$

$\ln x_1 = a$, $\ln \dfrac{1}{a} = a$, $-\ln a = a \Rightarrow a + \ln a = 0$

$g(x) = \ln x + x$

$g'(x) = \dfrac{1}{x} + 1$

$g(x)$ is increasing in $x \in (0, \infty)$

According to option $f\left(\dfrac{1}{e}\right) < 0$ & $f\left(\dfrac{1}{\sqrt{e}}\right) > 0$

$\therefore a \in \left(\dfrac{1}{e}, \dfrac{1}{\sqrt{e}}\right)$

55. $y = 3x + \alpha$

$C_1(2,2)$, $r_1 = 2$

Perpendicular distance from C_1 to line > 2

$|4 + \alpha| > 2\sqrt{10}$

$\alpha + 4 > 2\sqrt{10}$, $\alpha + 4 > -2\sqrt{10}$

$\alpha > 2\sqrt{10} - 4$, $\alpha < -2\sqrt{10} - 4$

$C_2(8,2)$, $r_2 = 3$

Perpendicular distance from C_2 to line > 3

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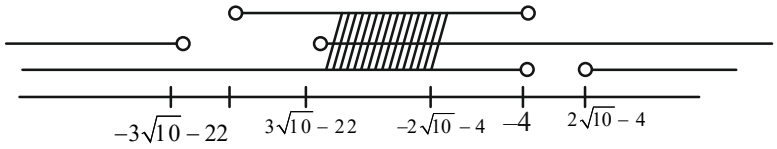
$\frac{|22+\alpha|}{\sqrt{10}} > 3$

$\alpha + 22 > 3\sqrt{10}, \alpha + 22 < -3\sqrt{10}$

$\alpha > 3\sqrt{10} - 22, \quad \alpha < -3\sqrt{10} - 22$

C_1 and C_2 lies on opposite side of $3x - y + \alpha = 0$

$(4+\alpha)(22+\alpha) < 0 \qquad \Rightarrow -22 < \alpha < -4$



$(3\sqrt{10} - 22, -2\sqrt{10} - 4)$

$a = 3, b = 22$

$c = -2, b = -4$

$a + b + c + d = 19$

56. $\int \frac{\text{cosec}^2 x - 2005}{\cos^{2005} x} dx$

$= \int (\cos x)^{-2005} \times \text{cosec}^2 x dx - 2005 \int \frac{dx}{\cos^{2005} x}$

$= (\cos x)^{-2005} \times (-\cot x) - \int (-2005)(\cos x)^{-2006} \times (-\sin x)$

$(-\cot x)dx - 2005 \int \frac{dx}{\cos^{2005} x}$

$= -\frac{\cot x}{(\cos x)^{2005}} + C$

57. $x^2(y+1)dx + y^2(x-1)dy = 0$

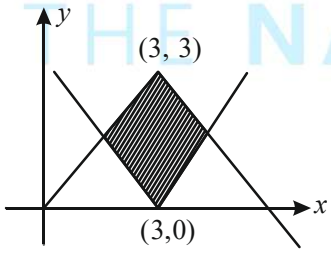
or $\frac{x^2 dx}{x-1} = -\frac{y^2 dy}{y+1}$

or $\int \left[x + 1 + \frac{1}{x-1} \right] dx = -\int \left[y - 1 + \frac{1}{y+1} \right] dy$

or $\frac{x^2}{2} + x + \ln(x-1) = -\left[\frac{y^2}{2} - y + \ln(y+1) \right] + \ln C$

$\Rightarrow x^2 + y^2 + 2(x-y) + 2\ln \frac{(x-1)(y+1)}{C} = 0$

58. Required area $= \frac{3}{\sqrt{2}} \times \frac{3}{\sqrt{2}} = \frac{9}{2}$



59. Put $y = x + 2$

60. Now we have to find x, y such that both are integers and $x^2 + y^2 = 64$

If $x = 0, y = \pm 8$

$y = 0, x = \pm 8$

$R = \{(0,8),(0,-8),(8,0),(-8,0)\}$

61. $\left[(2\vec{a} \times 3\vec{b}) \quad (3\vec{b} \times 4\vec{c}) \quad (4\vec{c} \times 2\vec{a}) \right]$

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$= 6\times12\times8\left[\vec{a}\times\vec{b}\quad\vec{b}\times\vec{c}\quad\vec{c}\times\vec{a}\right]$

$= 6\times12\times8\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]^2$

$= 6\times12\times8\times16 = 2^{10}\times3^2$

62. $x^2+3|x|-4 = 4\pi+14-4\pi$

$\Rightarrow x^2+3|x|-18=0$

$\Rightarrow |x|=3 \Rightarrow \cos^{-1}\cos 9 = 9-2\pi$

63. We have $4\cos^4 x - 2\cos 2x - \frac{1}{2}\cos 4x - x^7$

$= 4\cos^4 x - 2\left(2\cos^2 x-1\right) - \frac{1}{2}\left(2\cos^2 2x-1\right) - x^7$

$= 4\cos^4 x - 4\cos^2 x + 2 - \left(2\cos^2 x-1\right)2 + \frac{1}{2} - x^7$

$= \left(\frac{3}{2}-x^7\right)$

We get $g(x) = \left(\frac{3}{2}-x^7\right)^{\frac{1}{7}}$

$\Rightarrow g(g(x)) = \left(\frac{3}{2}-(g(x))^7\right)^{\frac{1}{7}}$

$= \left(\frac{3}{2}-\left(\frac{3}{2}-x^7\right)\right)^{\frac{1}{7}} = x$

Hence $g(g(100)) = 100$

64. Given curve is $\frac{x^2}{a-10}+\frac{y^2}{4-a}=1$

$(4-a)x^2+(a-10)y^2-(a-10)(4-a)=0$

Which represents an ellipse, if $h^2-ab<0$

$\Rightarrow -(4-a)(a-10)<0$

$\Rightarrow (a-4)(a-10)<0$

$\Rightarrow 4<a<10$

Hence the number of integral values of a is 5.

65. 1 B 2 B 3 B 4 B 5, 4! × (123 or 234 or 345) = 4!.3!

66. $\sum_{r=2}^{n}\frac{{}^{n-2}C_{r-2}}{\frac{n(n-1)}{r(r-1)}.{}^{n-2}C_{r-2}} = 20$

$\sum_{r=2}^{n}\frac{r(r-1)}{n(n-1)} = 20$

$\frac{1}{n(n-1)}\sum_{r=2}^{n}\left(r^2-r\right) = 20$

$\frac{1}{n(n-1)}\left(\frac{n(n+1)(2n+1)}{6}-1-\left(\frac{n(n+1)}{2}-1\right)\right) = 20$

$\Rightarrow n = 59$

67. The equation of any tangent to the parabola $y^2 = 8x$ is

$y = mx + \frac{2}{m}$ ……(i)

Since (i) is also a tangent of $xy=-1$, so $x\left(mx+\frac{2}{m}\right) = -1$

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$\Rightarrow \quad mx^2+\frac{2x}{m}+1=0$

$\Rightarrow \quad m^2x^2+2x+m=0$

Since it will provide us equal roots, so

$D=0$

$\Rightarrow 4-4m^3=0$

$\Rightarrow m^3=1$

$\Rightarrow m=1$

Hence, the equation of the common tangent is $y=x+2$.

68. Given question becomes

$$=\det\cdot A^{-1}\cdot\det\cdot\left(\text{adj}\left(\frac{\text{adj}\,B}{|B|}\right)\right)\cdot\det\cdot\left(\text{adj}\left(\frac{\text{adj}\,A}{|A|}\right)\right)$$

$$=\frac{1}{|A|}|\text{adj}(\text{adj}\,B)|\cdot|\text{adj}\,\text{adj}(-A)|$$

$$=\frac{1}{|A|}\cdot|B|^4\,(-1)\times|A|^4=-8$$

69. $p=\dfrac{{}^{13}C_2\cdot{}^4C_1\cdot{}^4C_1}{{}^{52}C_2}$ (probability that the cards are higher or lower rank)

$$=\frac{16}{17}$$

$\Rightarrow P(\text{same})=\dfrac{1}{17}$

Now $P(H)+P(L)+P(\text{same})=1$

$\therefore \quad P(H)=P(L)=\dfrac{8}{17}$

70. $2^{m-2n}e^{i\frac{m\pi}{6}}=e^{i\frac{n\pi}{4}}\Rightarrow 2^{m-2n}=e^{\left(\frac{n\pi}{4}-\frac{m\pi}{6}\right)}$

$$\Rightarrow\cos\left(\frac{n\pi}{4}-\frac{m\pi}{6}\right)=2^{m-2n}\ \&\ \sin\left(\frac{n\pi}{4}-\frac{m\pi}{6}\right)=0$$

$$\Rightarrow\cos\left(\frac{n\pi}{4}-\frac{m\pi}{6}\right)=2^{m-2n}=1$$

$$\Rightarrow m=2n\ \&\ \frac{m\pi}{6}=(n\pi/4)+2k\pi$$

$\Rightarrow n=24k$

$\therefore$ for least positive integer

$n=24$

$m=48$

71. Dividing both sides x^3

$$a+\frac{b}{x}+\frac{c}{x^2}+\frac{d}{x^3}=\begin{vmatrix}1-\frac{2}{x} & 1+\frac{1}{x} & 3\\ 2+\frac{1}{x} & 1-\frac{1}{x} & 2\\ 1-\frac{1}{x^2} & 1+\frac{2}{x} & 1\end{vmatrix}$$

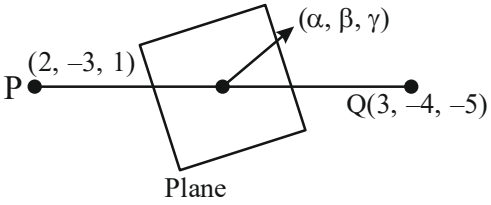
Now, $x\to\infty$

$$a+0+0+0=\begin{vmatrix}1 & 1 & 3\\ 2 & 1 & 2\\ 1 & 1 & 1\end{vmatrix}$$

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$$a=\begin{vmatrix}0 & 1 & 3\\ 1 & 1 & 2\\ 0 & 1 & 1\end{vmatrix}=-1(1-3)=2$$

72. Let α,β and γ are roots of $ax^3+bx^2+cx-d=0$



Let line PQ: $\frac{x-2}{1}=\frac{y+3}{-1}=\frac{z-1}{-6}=\lambda$

any point on line $(\lambda+2,-\lambda-3,-6\lambda+1)$

also lies on given plane $2x+y+z=7$

$\Rightarrow 2(\lambda+2)-\lambda-3-6\lambda+1=7$

$-5\lambda+2=7\Rightarrow\lambda=-1$

$\therefore$ hence point $(\alpha,\beta,\gamma)=(1,-2,7)$

$\alpha+\beta+\gamma=1-2+7=6\Rightarrow -b/a=6\ldots\ldots(1)$

$\sum\alpha\beta=-2-14+7\Rightarrow\frac{c}{a}=-9\ldots..(2)$

$\alpha\beta\gamma=(1)(-2)(7)\Rightarrow\frac{d}{a}=-14\ldots..(3)$

hence $\frac{-b+c-d}{a}$ put values of (1,) (2), (3)

$=6-9+14=11$

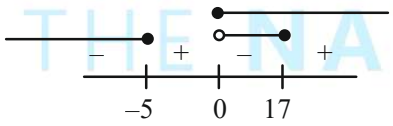
73. $\frac{\left(x^2+1\right)\left(x^2-12|x|-85\right)}{x}\le 0$

Case-I : $x\ge 0$ $\frac{\left(x^2+1\right)\left(x^2-12x-85\right)}{\pi}\le 0$

$\Rightarrow \frac{\left(x^2+1\right)\left(x^2-17x+5x-85\right)}{x}\le 0$

$\Rightarrow \frac{\left(x^2+1\right)(x(x-17)+5(x-17))}{x}\le 0$

$\Rightarrow \frac{(x^2+1)(x+5)(x-17)}{x}\le 0$



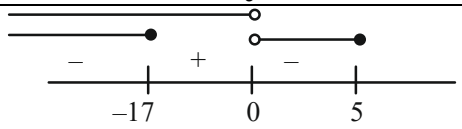
$\Rightarrow x\in(0,17]$

Case-II : $x<0$ $\frac{\left(x^2+1\right)\left(x^2+12x-85\right)}{x}\le 0$

$\frac{\left(x^2+1\right)\left(x^2+17x-5x-85\right)}{x}\le 0$

$\Rightarrow \frac{\left(x^2+1\right)(x+17)(x-5)}{x}\le 0$

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$x \in (-\infty, -17]$

$\therefore \quad x \in (-\infty, -17] \cup (0, 17]$

74. Given, $n(N) = 12, n(P) = 16, n(H) = 18, n(N \cup P \cup H) = 30$

From, $n(N \cup P \cup H) = n(N) + n(P) + n(H) - n(N \cap P)$

$-n(P \cap H) - n(N \cap H) + n(N \cap P \cap H)$

$n(N \cap P) + n(P \cap H) + n(N \cap H) = 16$

Now, number of pupils taking two subjects $= n(N \cap P) + n(P \cap H) + n(N \cap H) - 3n(N \cap P \cap H)$

$= 16 - 0 = 16$

75. $$h(x) = \begin{cases} |x^2 + 5x - 2| & , -8 \le x < 0 \\ \max(|x|, |x-2|), & 0 \le x < 2 \\ \min(|x-3|, 2), & 2 \le x < 6 \\ [2x]\operatorname{sgn}\left(x - \frac{15}{2}\right), & 6 \le x \le 8 \end{cases}$$

Solve, $[2x]\ \operatorname{sgn}\left(x - \frac{15}{2}\right)$

$-12 \quad , \quad 6 \le x < \frac{13}{2}$

$-13 \quad , \quad \frac{13}{2} \le x < 7$

$-14 \quad , \quad 7 \le x < \frac{15}{2}$

$0 \quad , \quad x = \frac{15}{2}$

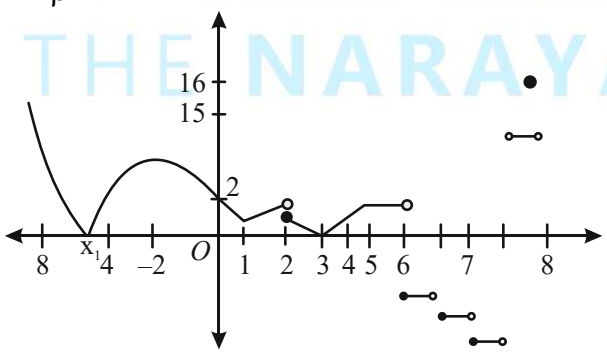
$15 \quad , \quad \frac{15}{2} < x < 8$

$16 \quad , \quad x = 8$

Discontinuous at $x = 2, 6, \frac{13}{2}, 7, \frac{15}{2}, 8$, $\therefore \alpha = 6$

Non-differentiable at $x = x_1, 0, 1, 2, 3, 5, 6, \frac{13}{2}, 7, \frac{15}{2}, 8$, ($x_1$ be a root of $x^2 + 5x - 2 = 0$ in $x < 0$)

$\therefore \quad \beta = 11$



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